ANALYSIS OF NYC 311 SERVICE REQUESTS IN 2025

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Repository: https://github.com/mayarareghin/nyc-311-sql-analysis

OBJECTIVE

- Analyze NYC 311 service requests to identify patterns in requests during 2025;
- Provide insights into the performance and operational efficiency of the 311 Service Request system;

TECHNOLOGIES

- **PYTHON**: Data ingestion and integration with DuckDB
- **SQL**: Data cleaning, transformation, and analysis
- **DUCKDB**: Analytical database for querying and processing data
- **PANDAS**: Data manipulation and dataset export
- **REST API**: Data extraction from NYC Open Data
- **GIT and GITHUB**: Version control

DATA ENGINEERING

The data used in this project refers to service requests made to the City of New York through the “311 Service Request” system in 2025 and was obtained via the NYC Open Data API.

The data ingestion process was implemented in Python using the NYC Open Data API. Due to the high volume of records (over 3 million), a parallel processing approach was adopted. Through parallelism, the code performs multiple simultaneous API requests instead of following a sequential flow of “request → wait → process → next request”. This significantly optimized data ingestion time.

As the data is processed, it is stored in Parquet format, partitioned by year. This format was chosen due to its query performance optimization, enabled by columnar storage (which allows reading only the necessary columns) and support for parallel processing, making it ideal for big data scenarios.

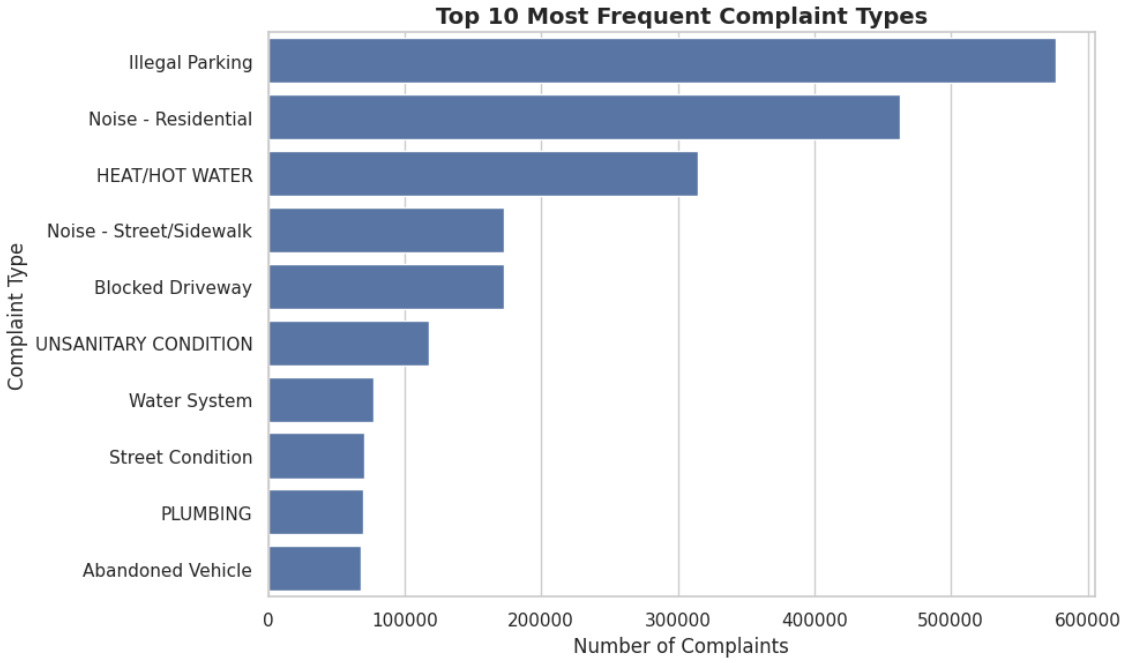
Data cleaning and transformation were performed using SQL. During the cleaning phase, columns deemed irrelevant to the analysis objective were removed, and a unified table was created.

After cleaning, new columns were created by converting date fields from VARCHAR to TIMESTAMP. Additional columns were derived to separate date and time components, facilitating analysis. A calculated column representing resolution time in hours (closure date – creation date) was also created.

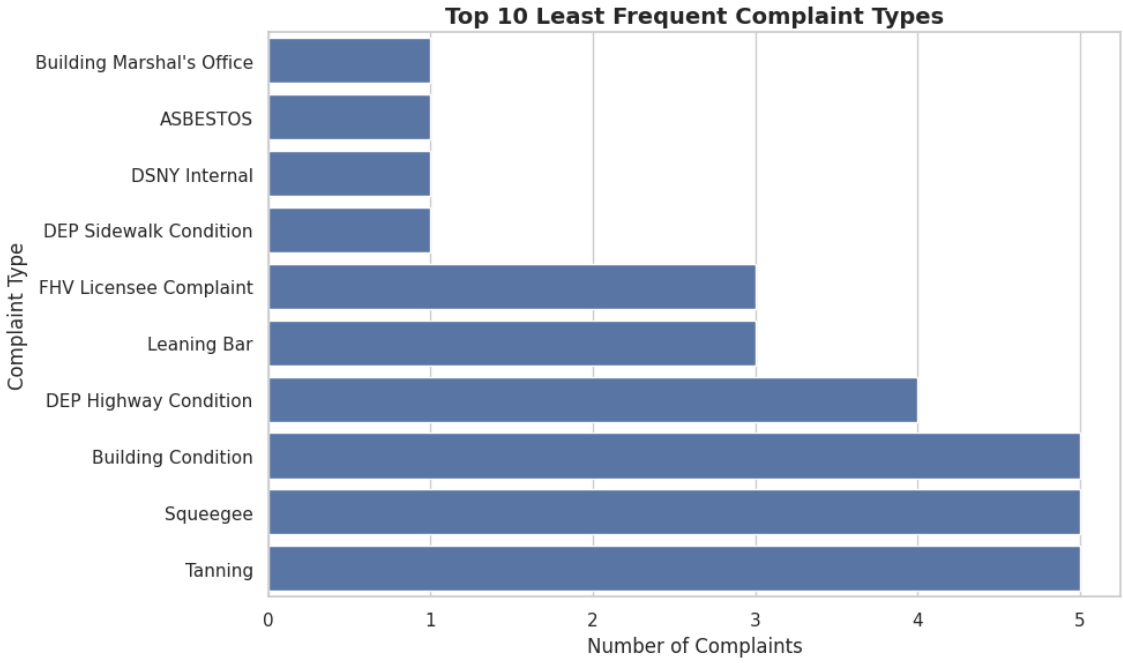
To execute these transformations and SQL queries, DuckDB was used. DuckDB is an open-source, serverless analytical database that allows SQL queries directly over files and performs efficiently with large datasets. It was integrated using Python.

DATA ANALYSIS

The analysis began by examining the types of complaints recorded in the dataset. A total of 191 distinct complaint types were identified. The top 10 most frequent complaints are primarily related to recurring urban issues such as noise, illegal parking, sanitation, and structural problems. This pattern indicates that most requests are related to everyday situations rather than isolated or exceptional events.

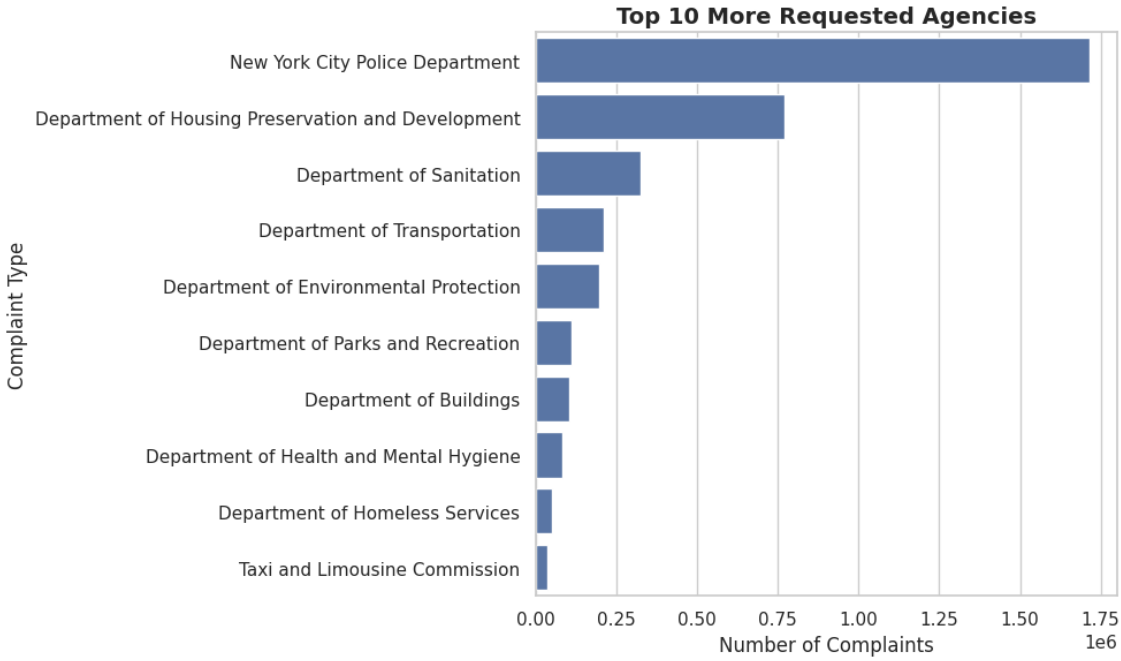


On the other hand, when analyzing the least frequent complaint types, categories with fewer than 5 records were observed, indicating potential outliers within the dataset.



Service requests in the 311 system are routed to different public agencies. The analysis shows that the police department accounts for approximately half of all requests, indicating a strong operational dependency on this agency. The most common complaints directed to this department involve non-emergency irregular situations such as excessive noise, illegal parking, and abandoned vehicles.

In contrast, only three agencies — the Office of Technology and Innovation, the Sheriff's Office, and the Department of Education — recorded fewer than 10,000 requests, indicating lower participation in the total request volume.



From a geographic perspective, among the five boroughs analyzed, Brooklyn has the highest number of complaints. This behavior can be partially explained by demographic factors, as it is the most populous borough in New York City.

Queens, Bronx, and Manhattan show relatively similar volumes, while Staten Island registers significantly fewer complaints, likely reflecting its lower population density and more residential profile.

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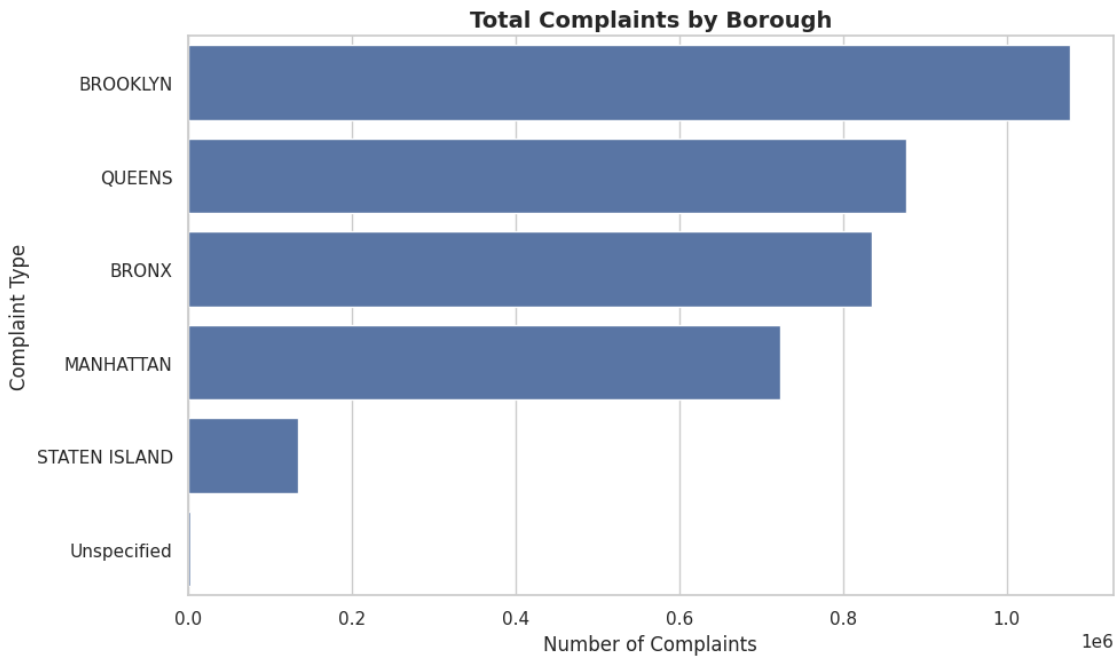

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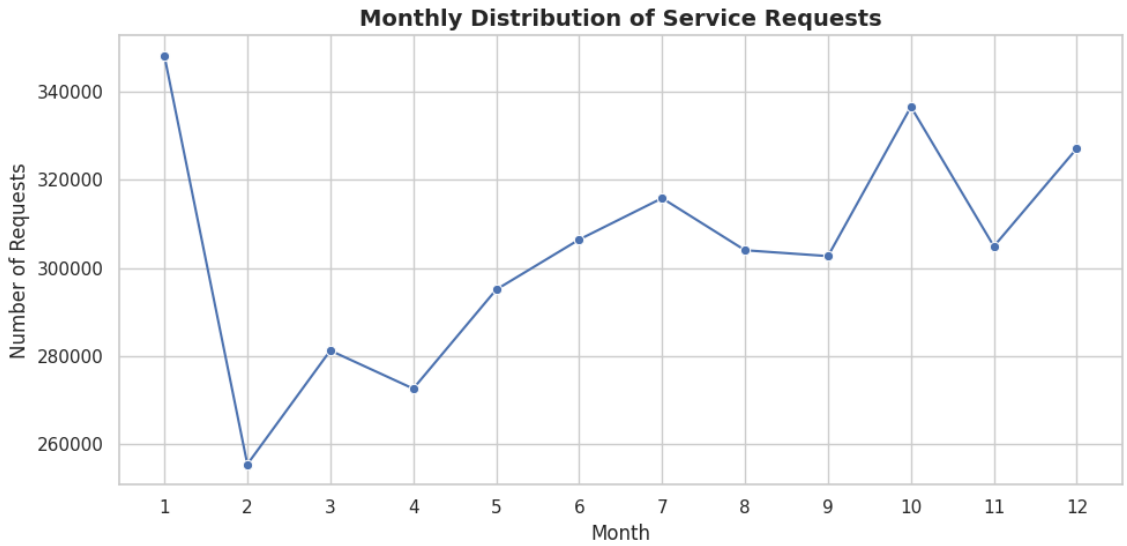
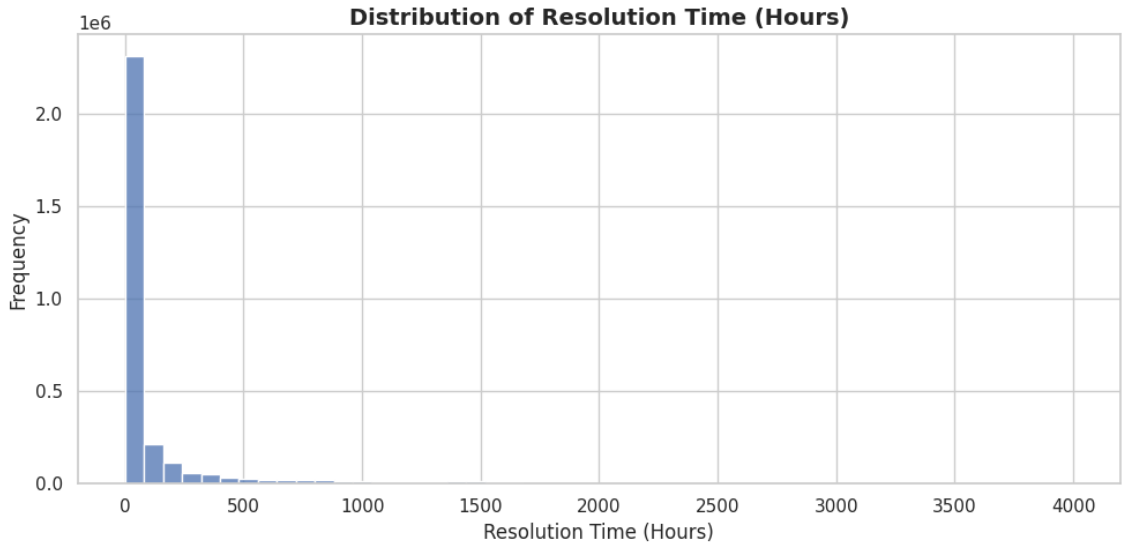


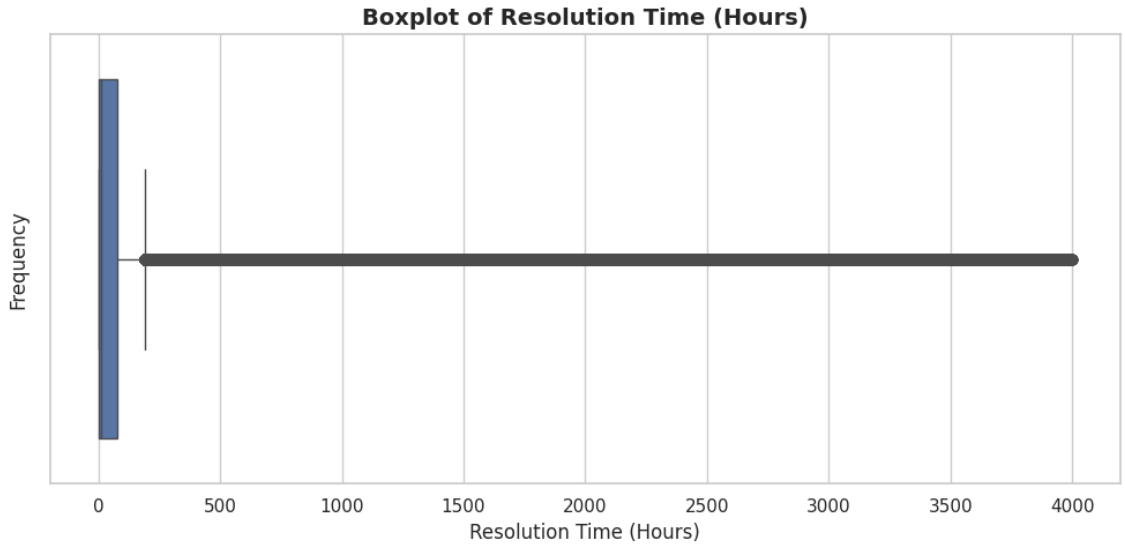
In terms of operational efficiency, approximately 98% of requests were resolved in 2025, indicating a high level of service performance.

Analyzing central tendency measures, the average resolution time is around 8 days, while the median is 6 hours. Minimum and maximum values indicate the presence of many

outliers. The minimum value (-4800 hours) corresponds to a data entry error, where a request created in 2025 was mistakenly recorded as closed in 2024. The maximum value corresponds to a request opened in 2025 and closed in 2026. Given these outliers, a new query was executed restricting resolution time between 0 hours (to remove negative values) and 8,766 hours (equivalent to one year).

Even after filtering, central tendency measures still indicate strong data asymmetry, which was confirmed through histogram and box-plot analysis. Despite a higher average and large maximum values, most cases have resolution times close to zero. This suggests that most requests are resolved within a few days, despite the presence of outliers.





However, a more detailed analysis reveals significant variation across different complaint types.

While the most frequent complaint types have an average resolution time of less than 2 days, some less frequent categories show extremely high resolution times, ranging from 93 to 267 days.

A deeper analysis shows that these cases are generally associated with very low occurrence volumes (often fewer than 25 records), suggesting that they are outliers rather than indicators of systemic inefficiency.

Thus, the results indicate that overall system performance is strongly driven by high efficiency in handling the most common requests, while a small set of rare and potentially more complex cases significantly impacts resolution time metrics if outliers are not treated.

INSIGHTS

Based on this analysis, some patterns can be observed:

• Complaints are mostly related to recurring urban issues

Most requests are associated with everyday problems such as noise, illegal parking, and sanitation issues, indicating that the 311 system is widely used to address recurring public demands.

• High concentration of requests in a single agency

The police department handles approximately half of all requests, highlighting a strong operational dependency on this agency.

• Geographic distribution aligned with population density

More populous boroughs, such as Brooklyn, have higher request volumes, while less dense areas, such as Staten Island, register significantly fewer requests.

• Stable request volume over time

The number of requests remains relatively constant throughout the months, indicating no significant seasonality and greater operational predictability.

• High resolution rate and operational efficiency

Approximately 98% of requests were resolved, demonstrating a high level of efficiency. The most common complaint types have an average resolution time of less than 2 days, indicating strong performance in handling recurring demands.